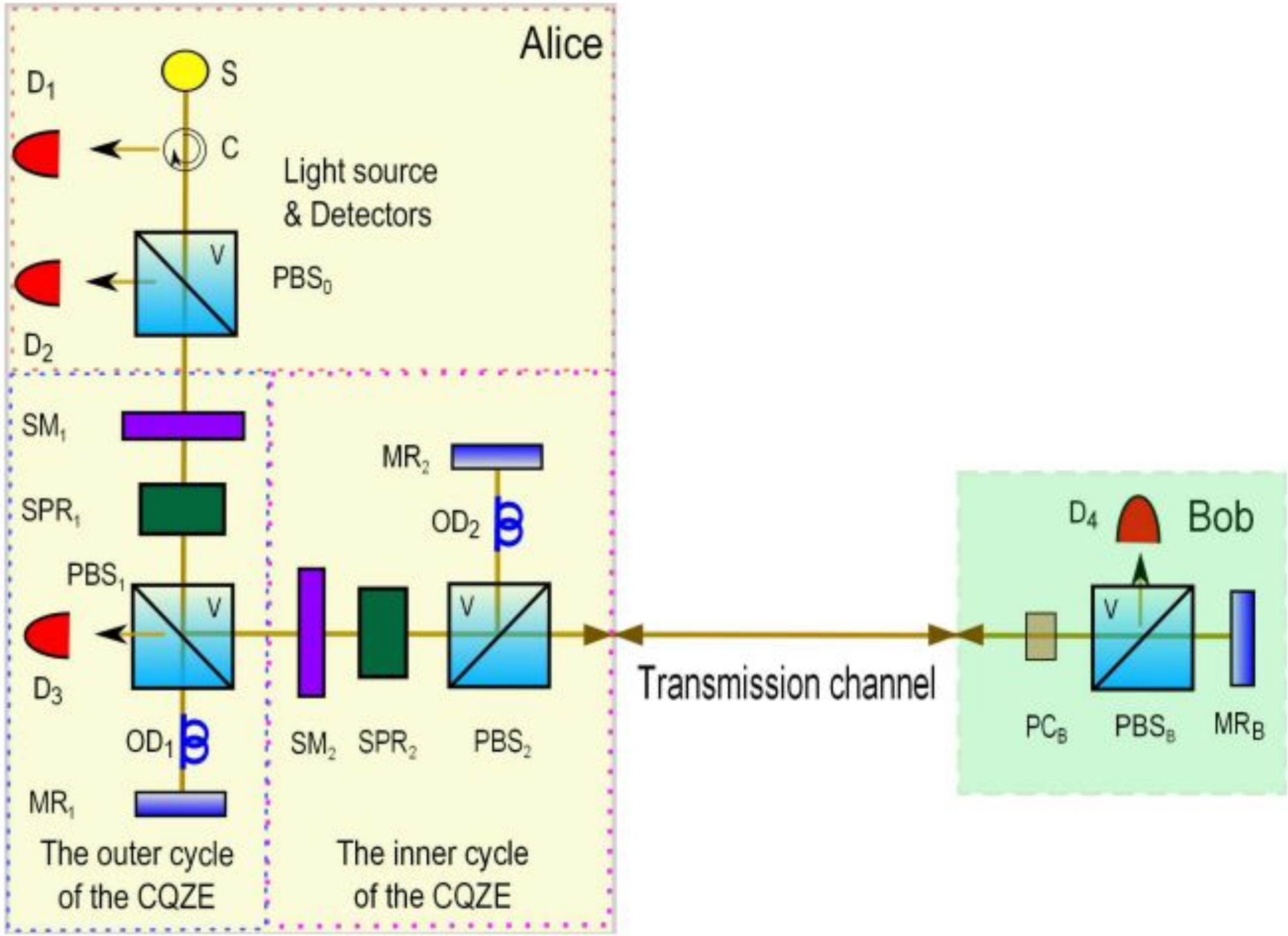
TEXAS A&M UNIVERSITY
Engineering

Qiskit implementation for Counterfactual Communication

Adhyantha Chandrasekaran

Core objective of simulation:
Computationally simulate the Chained Quantum Zeno Effect and demonstrate successful counterfactual communication

Main Idea:
Information is transmitted from Bob to Alice without physical particles traveling through the transmission channel



Qiskit AER Simulator

Nested Interferometers:

M = 4 Outer cycles

N = 4 Inner cycles

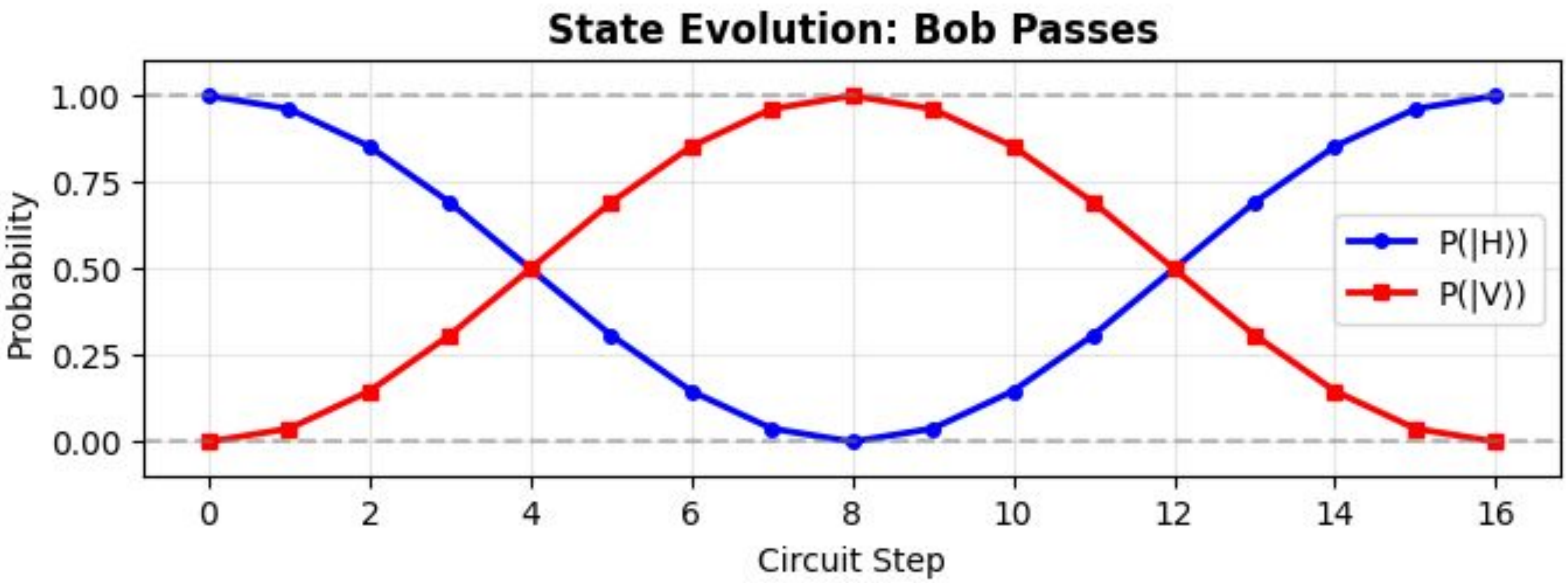
Gates:

Ry (rotation along y-axis):
simulates beam splitters modifying photon angle

Pauli X:
simulates Bob blocking path by reversing direction of photon

```
QISKIT QUANTUM CIRCUIT STRUCTURE
══════════════════════════════════════════════════════════

|0⟩=|H⟩ ── [RY]^4 ── [RY]^4 ── Bob's PC ── [RY†]^4 ── [RY†]^4 ── Measure
              |          |            |            |             |
        Outer Cycle   Inner Cycle   Transmission   Return Path     Detection
       (Alice's QZE) (Channel QZE)   (Bob acts)     (Reverse)     (D₁/D₂/D₃)

GATES USED:
- RY(θ): Rotation around Y-axis by angle θ
- X: Pauli-X gate (bit flip, polarization rotation H↔V)

PARAMETERS:
- M cycles (outer): 4
- N cycles (inner): 4
- Rotation angle θ_M: 0.1963 rad = 11.25°
- Rotation angle θ_N: 0.1963 rad = 11.25°
- Simulator: Qiskit Aer (statevector)

KEY RESULT:
✓ Alice detects Bob's bit without photon traveling channel!
✓ Quantum Zeno Effect prevents photon from entering channel
✓ Bob's action (PC on/off) changes final polarization state
```

Scenario 1: Bob passes



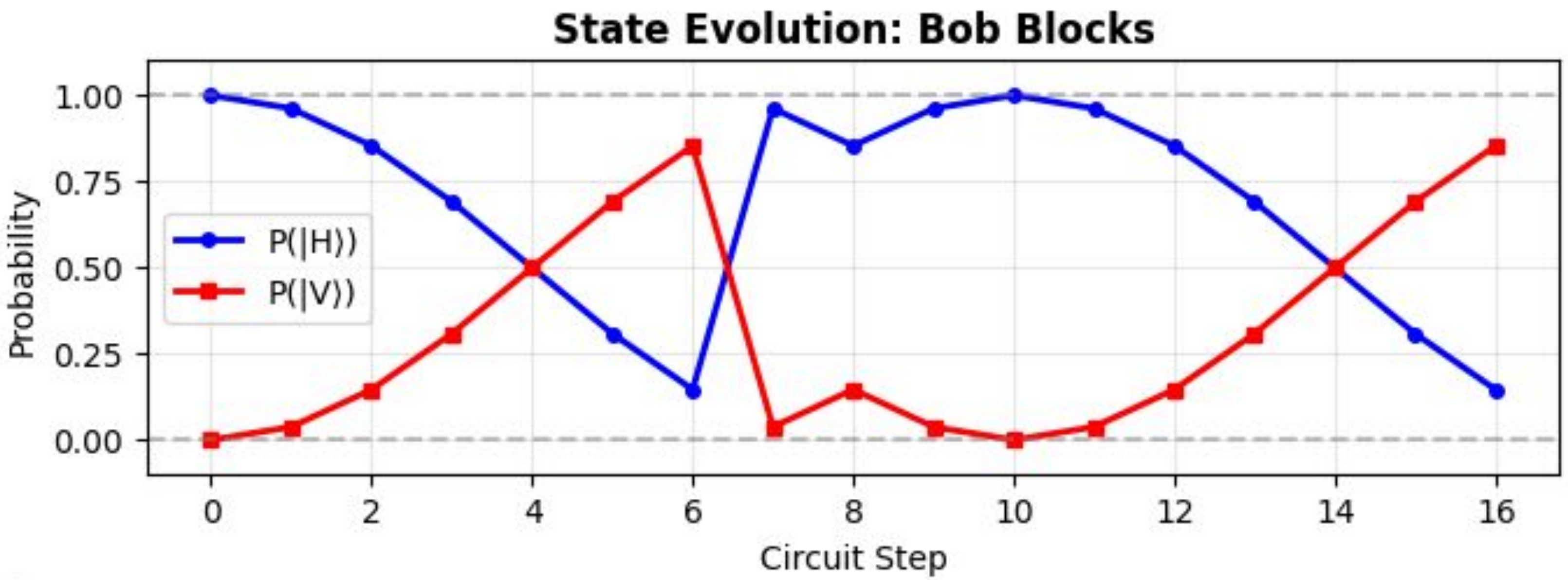
Bob’s bit = 0, no block

Simulation Result: Detector D3 (H = 0): 100.0%

Detectors D1/D2 (V = 1): 0.0%.

Conclusion: Perfect transmission of a '0' bit.

Scenario 2: Bob blocks



Bob's bit = 1, blocked

Simulation Result: Detector D3 (H = 0): 14.5% (depends on each trial)

Detectors D1/D2 (V = 1): 85.5%. (depends on each trial)

Conclusion: QZE successfully freezes evolution of a state

Counterfactual Proof

Channel leakage probability:

$$\boxed{P_{\text{leak}} = O\left(\frac{\pi^2}{16M^2N^2}\right)}$$

For M=N=4 : P_leak≈0.15%
For M=N=10 : P_leak≈0.024%
As M,N→∞ : P_leak→0

Yet Alice still learns Bob's bit! This is achieved through quantum interference, not photon transmission.

Quantum interference! NOT physical photon transmission

Summary

Base counterfactual Qiskit code is bug-free and validated against theory.

Next Step 1: Scale the Code. Increase M and N to minimize P_leak to near-zero.

Next Step 2: Advanced Measurements. Implement density matrix analysis and state fidelity metrics beyond basic bit counts.

Next Step 3: Quantum Ghost Imaging. Adapt this working CQZE framework to simulate ghost imaging protocols.

Source code: Feel free to suggest changes/contribute

Key Steps

1. **Initialize:** $|\psi_0\rangle = |0\rangle = |H\rangle$
2. **Outer cycle:** Apply M rotations $R_Y(2\theta_M)$
3. **Inner cycle:** Apply N rotations $R_Y(2\theta_N)$
4. **Bob acts:** Apply X (block) or I (pass) at mid-cycle
5. **Return:** Apply inverse rotations
6. **Measure:** Detect at D_1, D_2 (V) or D_3 (H)

Complete evolution:

$$\boxed{|\psi_{\text{out}}\rangle = R_Y(-2\theta_M)^M \cdot R_Y(-2\theta_N)^N \cdot U_{\text{Bob}} \cdot R_Y(2\theta_N)^N \cdot R_Y(2\theta_M)^M\,|H\rangle}$$

Success probabilities:

$$\begin{cases} P(H) \approx 1 & \text{Bob passes (bit=0)} \\ P(H) = P(V) \approx 0.5 & \text{Bob blocks (bit=1)} \end{cases}$$

Counterfactual limit:

$$\lim_{M,N\to\infty} P_{\text{leak}} = 0 \quad \text{while} \quad P_{\text{success}} \to 1$$

TEXAS A&M UNIVERSITY

Engineering